Karan Chawla Dora

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Experience

SAE AutoDrive - Toronto Autonomous Vehicle Team (aUToronto) — Sep. 2025 – Present
State Estimation Developer — *Toronto, ON*

- Developed a lane localization model to maintain vehicle positional accuracy during GPS outages, ensuring continuous and reliable autonomous navigation.
- Implemented an Iterated Extended Kalman Filter (**IEKF**) to process over 100 lane updates per 5 meters, optimizing real-time state estimation.
- Utilized **ROS** to process bag files, conduct data analysis, and validate state estimation algorithms across simulated and physical testing environments.

FIRST Robotics Competition — Jul. 2021 – Jun. 2025
Robotics Team Captain & Software Development Lead — *Ottawa, ON*

- Founding member and leader of a 90+ high school student team competing at a national level, qualifying for the **World Championship** and **2x Finalists at Provincial Championships**.
- Developed accurate autonomous routines in **C++**, leveraging **computer vision** and **odometry**.
- Enhanced robot performance and precision using feedforward and PID feedback **control systems**.

Research & Publications

Researcher @ Autonomous Space Robotics Lab (ASRL) | *ROS, RF, State Estimation* — May 2026 - Present

- Implementing a mesh radio network for long distance communication of a multi-robot fleet

Bruce Lab — University of Ottawa | *Research Assistant* — Aug. 2024

- Reduced transient time by 80% and steady-state error by 40% by implementing PID control systems in C++ to stabilize current and voltage in platinum-nickel electrodeposition for Fuel Cell Electric Vehicles.
- Reduced calibration time by one-third, developing a Python user interface to conduct trials **autonomously**.
- Expanded trial voltage limits by 5x, refining an **Arduino**-based potentiostat electric circuit.

Canadian Robotics and Artificial Intelligence Ethics Design Lab | *Robotics Research* — Aug. 2024

- Accelerated project timeline by 50% by characterizing DJI's Robomaster S1 in a simulative environment.
- Achieved 1-second target tracking by implementing computer vision models with Python **machine learning**.

Technical Skills

Languages: Python, C/C++, Java, Javascript, Typescript, HTML, CSS
Robotics & Control: ROS, PID control, State Estimation, Sensor Fusion
Tools & Libraries: NumPy, Matplotlib, pandas, Arduino, Git, Next.js

Awards

HackTheNorth 2024 2x Award Winner: Best Use of MongoDB, SymphonicLabs
2x FRC Provincial Finalists: FIRST Robotics 2023 , 2024
Harvard-MIT Math Tournament: Qualified Participant
FIRST Alumni Entrance Scholarship: Value of $15,000
Dean's List Semi-finalist: First Robotics Competition
University of Toronto Excellence Award (UTEA): For Summer Research Valued at $7,000

Education

University of Toronto — Toronto, ON
Bachelor of Applied Science — Engineering Science (cGPA: 3.95/4.0) — *2025 - 2030*

Research Interests: *Control systems, robotics autonomy, state estimation*